

Assignment #1

The zip file of Files4A1.zip contains three java files.

1. Under the project **Comp8490** created during installation, create a package in the name of **codes**849**, where ****** needs to be replaced by the initials of your first and last name.
2.
 - (a) In package **codes**849**, create three classes **A1.**.java**, **BaseShapes**.java**, and **Commons**.java** with ****** being your initials;
 - (b) Copy the content of the provided files to your corresponding classes;
 - (c) Run your **A1.**.java** file and make sure that a three-axis coordinate system and a white square box show up, with the string label adhered to one side of the box as illustrated in the left figure below.
3. Modify your java files to create a virtual world that simulates the main structure of a wind turbine with four primitive shapes, i.e., a cylinder for the tower, a sphere for the yaw drive, a box for the nacelle, and a thin box attached to a sphere for the rotor blades. All the four shapes need to be defined in **BaseShapes**.java** as derived classes of the abstract super class, and all the four objects use the same appearance definition in the Common's file but in different colors and have the **primflags** set to **Primitive.GENERATE_NORMALS**.
 - (a) Set the cylinder's radius to **0.12**, height to **1.0**, color to **Orange**, and resolution in both X and Y dimensions to **30**. While the cylinder's orientation is illustrated in the right figure below, its mass center coincides with the origin of the coordinate system. In addition, the top of the square box, which serves as the base for the wind turbine, touches the bottom surface of the cylinder.
 - (b) Set the sphere's color to **Red**, but keep its radius and resolution the same as the cylinder's setting so that exactly half of the sphere shows up when sitting its mass center on the top surface of the cylinder along the Y-axis.
 - (c) Set the box's width in Z-axis the same as the sphere's diameter, the height in Y-axis to **0.12**, and the length in X-axis to **0.52**. While the box sits right on top of the sphere, its mass center is shifted so that one end aligns with the sphere's boundary as illustrated in the figure.
 - (d) The rotor blades consist of a **Red** sphere in a radius of **0.06** and a **Magenta** box in the dimension of **0.01×0.06×0.5** in the X-, Y-, and Z-axis directions respectively. As illustrated in the figures below, the centers of the sphere and the box aligns with the nacelle's center. The box-shaped blades touch the sphere-shaped rotor, while the latter has only half visible as the other half is hidden inside the long end of the nacelle.
 - (e) Keep the whole structure moving while making the three-axis coordinate system stationary.
 - (f) Identify your work by moving the whitened string onto the side of the box and making it visible when the box's long end points to the left, which is illustrated in the right figure below. In addition, both the string label and the frame's title need to be updated and adequate amount of comments are required to make your program easy to comprehend.
4. Locate the folder that contains all files of your project **Comp8490**; produce a zip file of your project folder; and submit it online before due.

